University of Pittsburgh

CS 2410 Introduction

Technology Constraints

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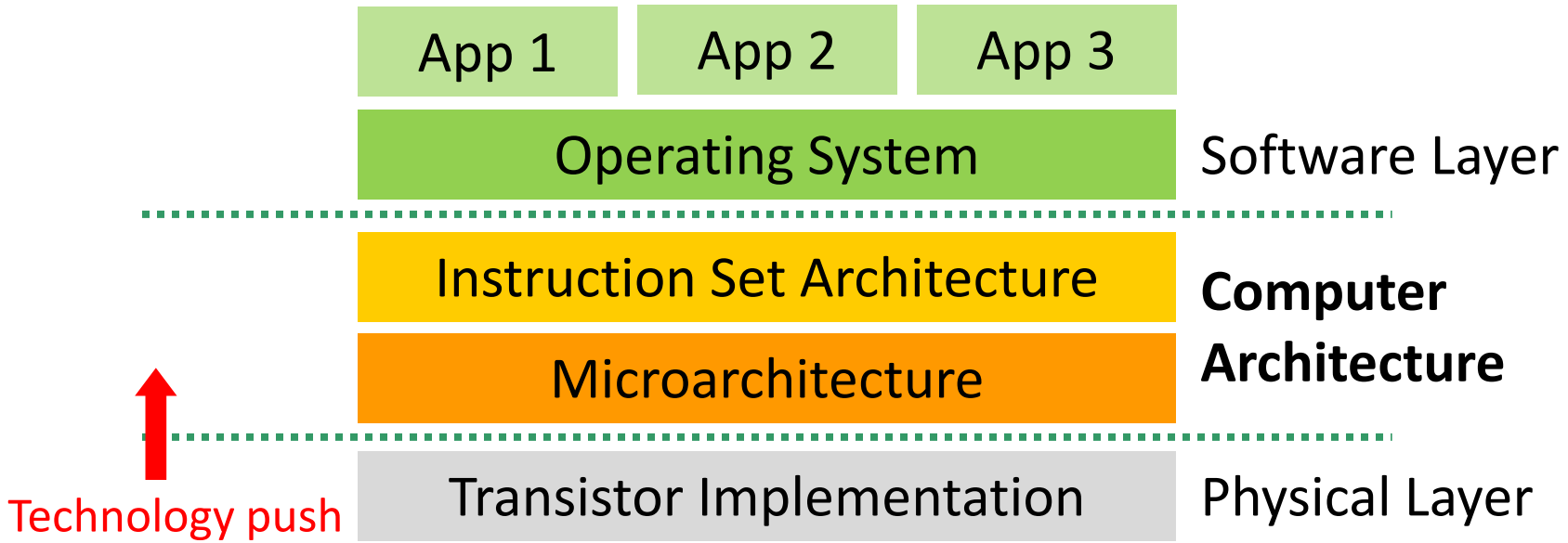
- Constraints in technology push architecture too



  - *Power Wall*: Thermal Design Power (TDP) constraint
  - *Memory Wall*: Constraint in bandwidth to memory
  - *Variability*: Limits in the precision of manufacturing technology
- Processor must be designed to meet all constraints

Power Wall

- $Power_{CPU} = Power_{dynamic} + Power_{leakage}$
  $Power_{dynamic} \propto A * N * CFV^2$
  $Power_{leakage} \propto f(N, V, V_{th}) \propto N * V * e^{-Vth}$
  - Leakage power is also called *static power*
  - This total CPU power cannot exceed TDP

- Moore's Law transistor scaling means two things:
  - N = Number of transistors ($\propto$ 1/transistor size^2) ⇧ ⇧
  - C = Capacitance ($\propto$ transistor size) ⇩
  - Reductions in C does not compensate for increases in N

- Architects must use tricks to keep power in check
  - To keep packing more transistors to increase performance

1. Reducing Dynamic Power

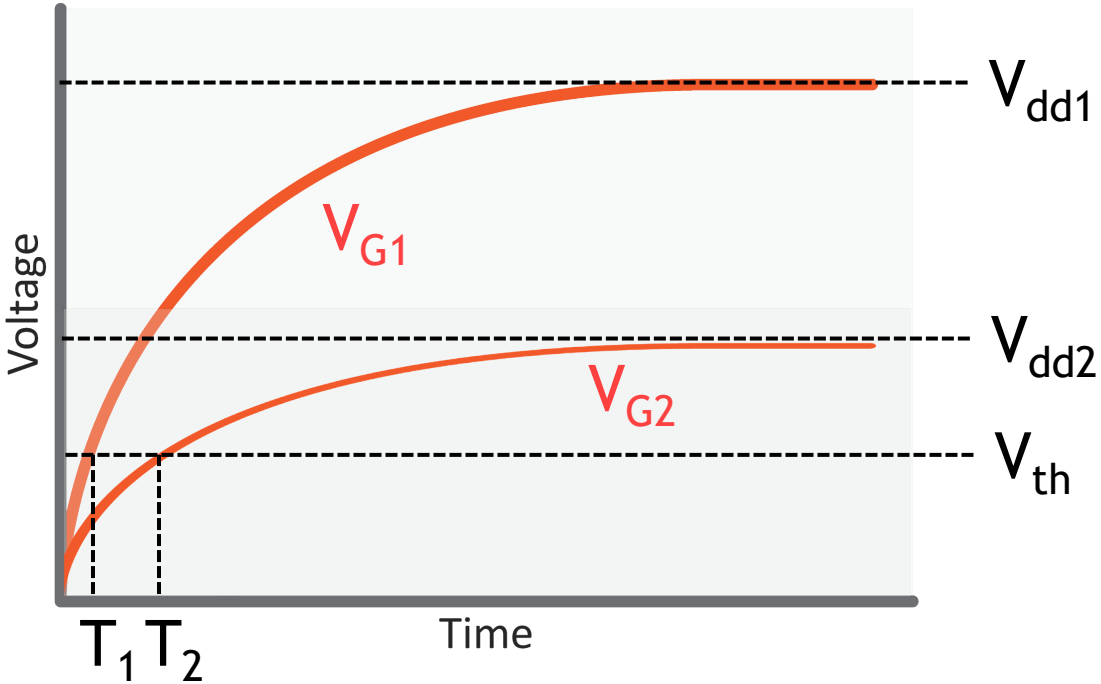
- $Power_{dynamic}$ ($\propto A * N * CFV^2$) + $Power_{leakage}$ ($\propto N * V * e^{-Vth}$)

- Reducing A (Activity): *Clock gating*
  - Disables the clock signal to unused parts of the chip (idle cores)
  - Wake-up is instantaneous (the moment clock signal goes in)

- Reducing F (Frequency) and V (Supply Voltage)
  - When F is reduced, V can also be reduced
    (The RC curve and water pressure, remember?)
  - *Dynamic Voltage Frequency Scaling (DVFS)* done on multi-cores
    - Slow down low-priority cores, speed up high-priority cores

DVFS and Transistor Speed

- RC Charging Curve of V_G



- Delay (T) increases with supply voltage (V_{dd}) decrease → Frequency (1/T) decreases with V_{dd} decrease

2. Reducing Leakage Power

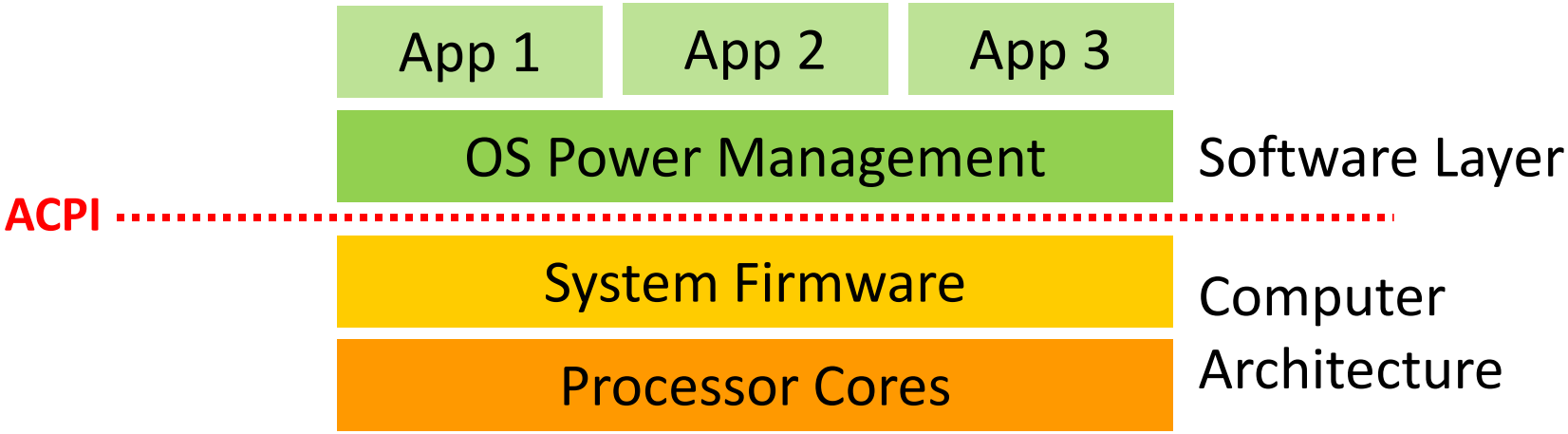
- $Power_{dynamic}$ ($\propto A * N * CFV^2$) + $Power_{leakage}$ ($\propto N * V * e^{-Vth}$)

- Reducing N (Transistor Number): *Power gating*
  - Disables power to unused parts of the chip (unused cores)
  - Eliminates dynamic power *and* leakage power to those parts
  - Drawback: wake-up takes a much longer time than clock gating
    - Delay for supply voltage to stabilize
    - Delay to backup and restore CPU state to/from memory

- Reducing V (Supply Voltage): *DVFS* also helps here
  - Only the voltage scaling part, not the frequency scaling

OS Manages Power

- Who decides which cores to clock gate and power gate?
- Who decides how to apply DVFS to the cores?
- *ACPI* (*Advanced Configuration and Power Interface*)
  - OS performs power management using this interface
    - OS knows best which threads to prioritize for best user experience
  - Open standard interface to system firmware
    - Firmware sends signals to processor cores to control them



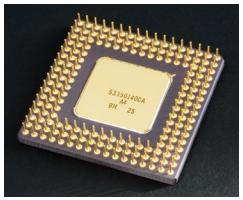
3. Simpler Processor Design

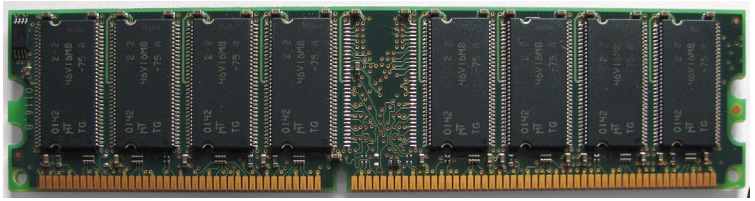
- Plenty of transistors but not enough power
  - Power becomes the ultimate currency in processor design
- Complex logic for performance is power hungry
  - Not easy to eke out the last bit of performance out of a program
  - Diminishing returns on performance for power investment
- Push towards simpler architectures:
  - *Multi-cores*: Run multiple programs (threads) on simple cores
  - *GPUs*: Run each instruction on massively parallel compute units
  - *Caches*: Memory caches are power efficient (low dynamic power)

Memory Wall

- Refers to both *latency (ns)* and *bandwidth (GB/s)*
  - CPU frequency and overall performance increased dramatically
  - Memory (DRAM) latency and bandwidth have lagged far behind

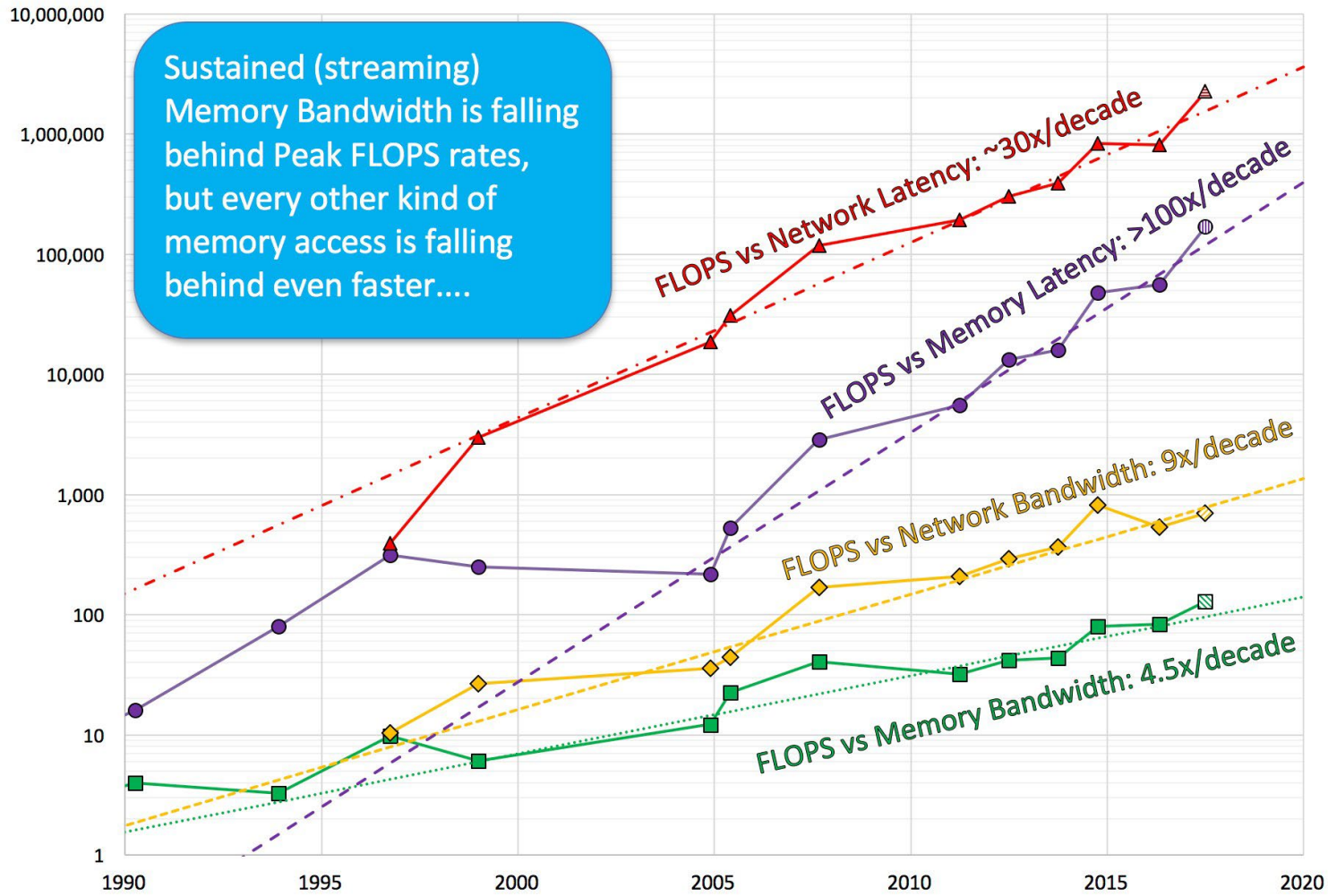
- Why?
  - Limit on the number of CPU / DRAM pins that can be soldered on





  - DRAM manufacturers have traditionally prioritized capacity
  - DDR1 (1998): 1.6 GB/s → DDR4 (2014): 25.6 GB/s
    (Impressive? Not so much compared to CPU performance)

Memory Wall

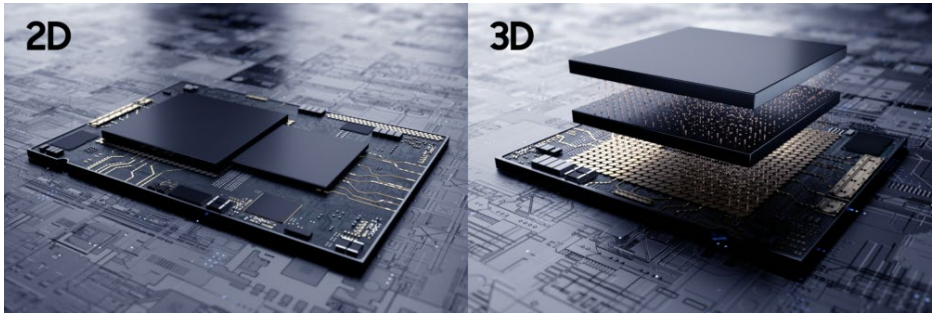


Source: SC16 Invited Talk ""Memory Bandwidth and System Balance in HPC Systems" by John D. McCalpin

- FLOPS = floating point operations per second (performance)

Memory Wall

- Where did the Memory Wall push architecture?
- *Caches*: If hit in cache, no need to go to memory
  - Caching reduces both data access latency/bandwidth
- *3D-Stacked Memory:* Stack CPU on top of memory
  - Drill vias, or holes, through silicon to bond CPU with memory
  - Through silicon vias (TSVs) have low latency / high bandwidth

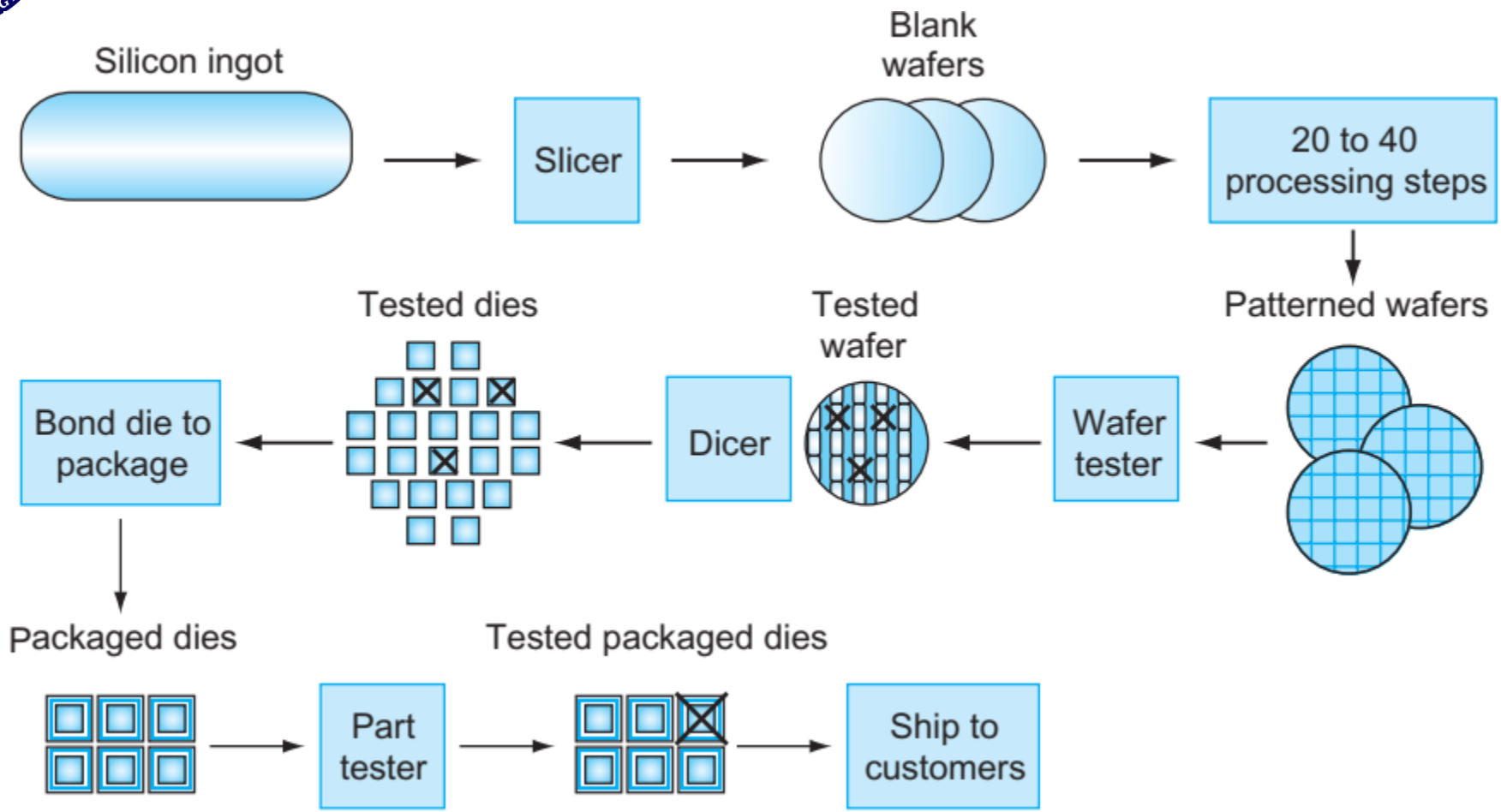


Variability

- *Variability*: differences in speed of individual transistors
  - If fab can't ensure uniformity of transistors, speeds will differ
  - Speed differences mostly come from variations in V_{th}:
    low V_{th} → cycle time ⇩ but leakage power ⇧
    high V_{th} → cycle time ⇧ but leakage power ⇩

- If unlucky and a logic path has lots of *slow* transistors
  → CPU may miss clock cycle time if path is exercised
  → CPU must be discarded, since it malfunctions
- If unlucky and a region has too many *fast* transistors
  → Region may generate too much heat due to low V_{th}
  → CPU must be discarded, due to overheating
- Leads to low chip *yield*

Wafer Yield



- Lower yield leads to higher production cost

Variability

- Where did Variability push architecture?

- *Product binning*: Sell slower CPUs at a cheaper "bin"
  - And rate slower CPUs at a lower CPU frequency
  - Instead of discarding them as "malfunctioning"
- *Multi-cores*: Easy to disable one or two buggy cores
  - Compared to single core where subcomponents must be disabled
  - When cores are disabled, sold as a cheaper chip
- *Limited pipelining*: pipelining exacerbates variability
  - With long stages, many transistors so tend to even each other out
  - With short stages, few transistors so probable all are slow

Intel i9 Product Binning

Model	# Cores	# Threads	Base Clock	All Core Turbo	Turbo Boost	Total L3 Cache	PL1 TDP
i9-10900K	10	20	3.7	4.8	5.1	20	125
i9-10900KF	10	20	3.7	4.8	5.1	20	125
i9-10900	10	20	2.8	4.5	5.0	20	65
i9-10900F	10	20	2.8	4.5	5.0	20	65
i9-10900T	10	20	1.9	3.7	4.5	20	35
i7-10700K	8	16	3.8	4.7	5.0	16	125
i7-10700KF	8	16	3.8	4.7	5.0	16	125
i7-10700	8	16	2.9	4.6	7.7	16	65
i7-10700F	8	16	2.9	4.6	4.7	16	65
i7-10700T	8	16	2.0	3.7	4.4	16	35
i5-10600K	6	12	4.1	4.5	4.8	12	125
i5-10600K	6	12	4.1	4.5	4.8	12	125
i5-10600	6	12	3.3	4.4	4.8	12	65
i5-10600T	6	12	2.4	3.7	4.0	12	35
i5-10500	6	12	3.1	4.2	4.5	12	65
i5-10500T	6	12	2.3	3.5	3.8	12	35
i5-10400	6	12	2.9	4.0	4.3	12	65
i5-10400F	6	12	2.9	4.0	4.3	12	65
i5-10400T	6	12	2.0	3.2	3.6	12	35

Why the close to 4X difference? Clock difference is just 2X!

Produced from one wafer

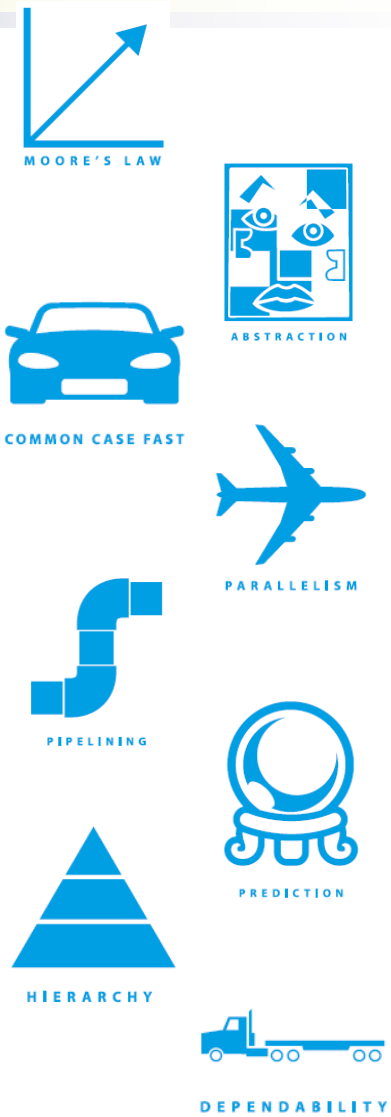
4 out of 10 cores were discarded.

Source: https://www.techspot.com/article/2039-chip-binning/

* TDP is calculated using the Base Clock frequency at a nominal supply voltage

Opportunities for Speed Improvement

- So Dennard Scaling is dead
  - Free CPU frequency gains are no longer there
- And we are walled in by technology constraints
  - Power wall
  - Memory wall
  - Variability
  - …
- Where do architects go look for performance?
  - Remember, Moore's Law is still giving us tons of transistors!

8 Great Ideas in Computer Architecture

- Design for *Moore's Law*
- Use *abstraction* to simplify design
- Make the *common case fast*
- Performance *via parallelism*
- Performance *via pipelining*
- Performance *via prediction*
- *Hierarchy* of memories
- *Dependability via* redundancy